

Motion in a Plane

SHORT NOTES

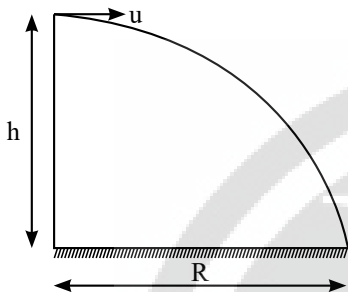
PROJECTION MOTION

Horizontal Projection from a Height h

Time of flight: $T = \sqrt{\frac{2h}{g}}$

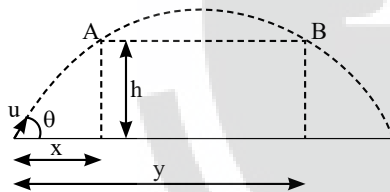
Horizontal range: $R = uT = u\sqrt{\frac{2h}{g}}$

Angle of velocity at any instant with horizontal: $\theta = \tan^{-1}\left(\frac{gt}{u}\right)$



Projectile Motion

- ❖ A body crosses two points at same height in time t_1 and t_2 , the points are at distance x and y from starting point then



- + $x + y = R$
- + $t_1 + t_2 = T$
- + $h = \frac{1}{2}gt_1t_2$

Velocity of Particle at Time t during projectile

- ❖ $\vec{v} = v_x\hat{i} + v_y\hat{j} = u_x\hat{i} + (u_y - gt)\hat{j} = u\cos\theta\hat{i} + (u\sin\theta - gt)\hat{j}$
- ❖ If angle of velocity $\vec{v}$ from horizontal is α , then

$$\tan \alpha = \frac{v_y}{v_x} = \frac{u_y - gt}{u_x} = \frac{u\sin\theta - gt}{u\cos\theta} = \tan\theta - \frac{gt}{u\cos\theta}$$

At highest point: $v_y = 0, v_x = u\cos\theta$

Time of flight: $T = \frac{2u_y}{g} = \frac{2u\sin\theta}{g}$

Horizontal range:

$$R = (u\cos\theta)T = \frac{2u^2\sin\theta\cos\theta}{g} = \frac{u^2\sin 2\theta}{g} = \frac{2u_xu_y}{g}$$

It is same for θ and $(90^\circ - \theta)$ and maximum for $\theta = 45^\circ$.

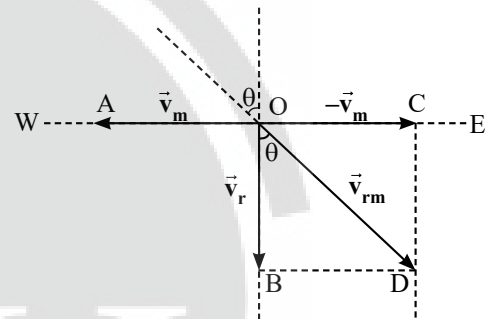
Maximum height: $H = \frac{u_y^2}{2g} = \frac{u^2\sin^2\theta}{2g} = \frac{1}{8}gT^2, \frac{H}{R} = \frac{1}{4}\tan\theta$

Equation of trajectory:

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta} = x\tan\theta\left(1 - \frac{x}{R}\right)$$

RELATIVE MOTION

Relative Velocity of Rain w.r.t. the Moving Man



The relative velocity of rain w.r.t. man $\vec{v}_{rm} = \vec{v}_r - \vec{v}_m$ will be represented by diagonal $\overline{OD}$

$$\therefore v_{rm} = \sqrt{v_r^2 + v_m^2 + 2v_rv_m\cos 90^\circ} = \sqrt{v_r^2 + v_m^2}$$

If θ is the angle which $\vec{v}_{rm}$ makes with the vertical direction then

$$\tan\theta = \frac{BD}{OB} = \frac{v_m}{v_r} \Rightarrow \theta = \tan^{-1}\left(\frac{v_m}{v_r}\right)$$

Swimming into the River

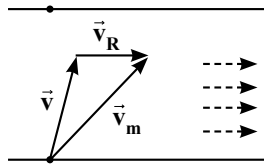
- + If the swimming is in the direction of flow of water or along the downstream then

$$\vec{v} \quad \vec{v}_R \quad v_m = v + v_R$$

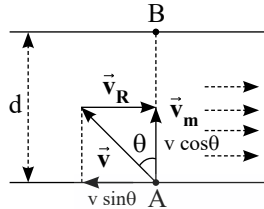
- + If the swimming is in the direction opposite to the flow of water or along the upstream then

$$\vec{v} \quad \vec{v}_R \quad v_m = v - v_R$$

- ✦ If man is crossing the river as shown in the figure *i.e.* $\vec{v}$ and $\vec{v}_R$ are non-collinear then use the vector algebra
 $\vec{v}_m = \vec{v} + \vec{v}_R$ (assuming $v > v_R$)



For shortest path

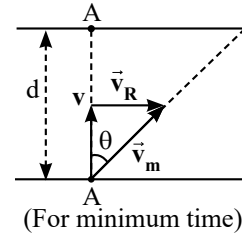


(For minimum displacement)

- ✦ To reach at B, $v \sin \theta = v_R \Rightarrow \sin \theta = \frac{v_R}{v}$

- ✦ Time of crossing, $t = \frac{d \cos \theta}{v}$

For minimum time



then, Drift = $t_{\min} v_R = \frac{d}{v} (v_R)$